animal (all geneticists think it is) then it must also be legitimate to speak of a gene as affecting the hardness of the stones in a caddis house. Startling thought, isn't it? Yet the reasoning is inescapable.

We are ready for the next step in the argument: genes in one organism can have extended phenotypic effects on the body of another organism. Caddis houses helped us take the previous step; snail shells will help us take this one. The shell plays the same role for a snail as the stone house does for a caddis larva. It is secreted by the snail's own cells, so a conventional geneticist would be happy to speak of genes 'for' shell qualities such as shell thickness. But it turns out that snails parasitized by certain kinds of fluke (flatworm) have extra-thick shells. What can this thickening mean? If the parasitized snails had had extra-thin shells, we'd happily explain this as an obvious debilitating effect on the snail's constitution. But a *thicker* shell? A thicker shell presumably protects the snail better. It looks as though the parasites are actually helping their host by improving its shell. But are they?

We have to think more carefully. If thicker shells are really better for the snail, why don't they have them anyway? The answer probably lies in economics. Making a shell is costly for a snail. It requires energy. It requires calcium and other chemicals that have to be extracted from hard-won food. All these resources, if they were not spent on making shell substance, could be spent on something else such as making more offspring. A snail that spends lots of resources on making an extra-thick shell has bought safety for its own body. But at what cost? It may live longer, but it will be less successful at reproducing and may fail to pass on its genes. Among the genes that fail to be passed on will be the genes for making extra-thick shells. In other words, it is possible for a shell to be too thick as well as (more obviously) too thin. So, when a fluke makes a snail secrete an extra-thick shell, the fluke is not doing the snail a good turn unless the fluke is bearing the economic cost of thickening the shell. And we can safely bet that it isn't being so generous. The fluke is exerting some hidden chemical influence on the snail that forces the snail to shift away from its own 'preferred' thickness of shell. It may be prolonging the snail's life. But it is not helping the snail's genes.

What is in it for the fluke? Why does it do it? My conjecture is the following. Both snail genes and fluke genes stand to gain from the snail's bodily survival, all other things being equal. But survival is not the same thing as reproduction and there is likely to be a trade-off. Whereas snail